# Permutations

Name _____

**The counting principle.** If one choice has $m$ options and a second choice has $n$ options, there are $m \cdot n$ ways to make both choices.

**Example:** 3 shirts and 4 pairs of pants make $3 \cdot 4 = 12$ outfits.

**Arranging.** In how many ways can 4 students sit in 4 chairs in a row?

| Chair 1 | Chair 2 | Chair 3 | Chair 4 |
|---|---|---|---|
| 4 choices | 3 left | 2 left | 1 left |

$4 \cdot 3 \cdot 2 \cdot 1 = 24$ ways

**Factorial:** $n! = n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1$ read "$n$ factorial"

$4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$ $1! = 1$ $0! = 1$ (there is exactly one way to arrange nothing)

**Permutations.** A **permutation** is an arrangement where **order matters**.

**Example:** A club of 10 students picks a president, a vice president, and a secretary. How many ways? $10 \cdot 9 \cdot 8 = 720$ ways. (Ana as president and Ben as vice president is different from Ben as president and Ana as vice president.)

**Permutation formula:** $${}_nP_r = \frac{n!}{(n-r)!}$$ arrange $r$ things chosen from $n$ things

$${}_{10}P_3 = \frac{10!}{7!} = \frac{10 \cdot 9 \cdot 8 \cdot \cancel{7 \cdot 6 \cdots 1}}{\cancel{7 \cdot 6 \cdots 1}} = 720$$

${}_nP_r$ is read "$n$ permute $r$." Some books write $P(n, r)$.

**Repeats allowed?** 3-letter codes from 26 letters:

No repeats: $26 \cdot 25 \cdot 24 = 15{,}600$ (a permutation)

Repeats allowed: $26 \cdot 26 \cdot 26 = 26^3 = 17{,}576$ (not a permutation; use the counting principle)

Work with a partner. Compare answers when you finish each problem.

1. Find each value.

   (a) $5!$

   (b) $\dfrac{6!}{4!}$

   (c) $0!$

   (d) ${}_8P_3$

2. You have 5 shirts, 3 pairs of pants, and 2 pairs of shoes. How many outfits can you make?

3. In how many ways can 6 different books be arranged on a shelf?

4. 8 runners are in a race. In how many ways can gold, silver, and bronze medals be given?

5. A phone code has 4 digits (0 to 9).

   (a) How many codes are there if digits can repeat?

   (b) How many codes are there if no digit repeats?

6. In how many ways can you arrange the letters of the word MATH?

7. A club of 12 members picks a president and a treasurer. One person cannot have both jobs. How many ways?

8. Explain why ${}_{10}P_3$ is the same as $10 \cdot 9 \cdot 8$.